1) $$\begin{array}{cc} c_1 & c_2 \end{array}\quad \vec{v}$$
$$\begin{bmatrix}1&2\\3&4\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = 2\begin{bmatrix}1\\3\end{bmatrix}+1\begin{bmatrix}2\\4\end{bmatrix}=\begin{bmatrix}2\\6\end{bmatrix}+\begin{bmatrix}2\\4\end{bmatrix}=\begin{bmatrix}4\\10\end{bmatrix}$$

2) $$a\begin{bmatrix}1\\2\end{bmatrix}+b\begin{bmatrix}2\\1\end{bmatrix}=\begin{bmatrix}5\\7\end{bmatrix} \Rightarrow \begin{matrix} a+2b=5 & (1) \\ 2a+b=7 & (2)\end{matrix} \Rightarrow$$

(1) $a=5-2b$

(2) $b=7-2a \Rightarrow 7-2(5-2b)=b \Rightarrow 7-10-4b=b \Rightarrow$

$-3=3b \therefore b=-1$

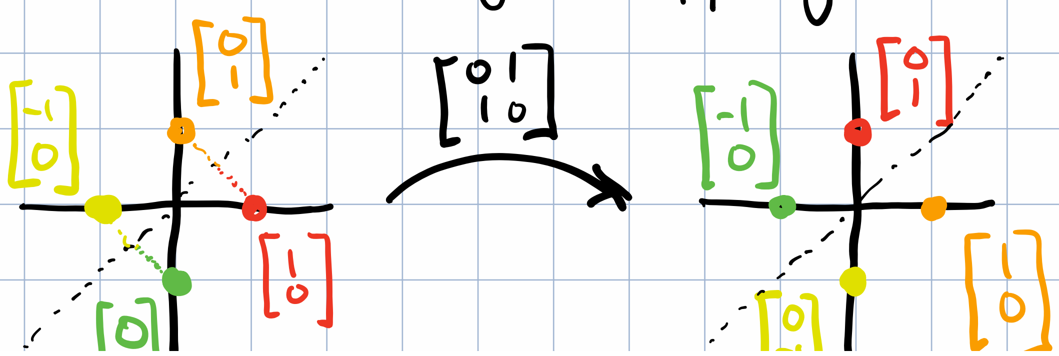
$a=5-2=3$

3) $\begin{bmatrix}3&0\\0&3\end{bmatrix}=3I$ The matrix scales vectors by 3.

4) $\begin{bmatrix}0&1\\1&0\end{bmatrix}$ reflects a vector along the line $y=x$

$$\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=x\begin{bmatrix}0\\1\end{bmatrix}+y\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}0\\x\end{bmatrix}+\begin{bmatrix}y\\0\end{bmatrix}=\begin{bmatrix}y\\x\end{bmatrix}$$

It is essentially swapping x and y



5) The human mind has an intuitive sense of geometry. Thinking of matricies as transformations on vectors gives us an idea of how the matrix acts on the vector much better than memorizing a table of numbers and blindly following the arithmatic.